HAOMING WANG

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RESEARCH INTERESTS

Video Benchmark Synthesis & Evaluation, Diffusion / Generative Image & Video Models, Latent-Representation Shaping for VLMs, Cross-Frame Temporal Consistency.

EDUCATION

Sept 2022 – May 2027 (Anticipated) **Ph.D. in Electrical and Computer Engineering**
University of Pittsburgh, Pittsburgh, PA, USA
Advisor: Prof. Wei Gao, Intelligent System Lab

Sept 2018 – May 2022 **B.Eng. in Automation** GPA: 3.8/4.0
Zhejiang University, Hangzhou, China
Chu Kochen Honors College

EXPERIENCE

Sept 2022 – Present **Graduate Student Researcher**
Intelligent System Lab, University of Pittsburgh, Pittsburgh, PA, USA

- **(May 2025 – Present)** Led research on procedural / LLM-agentic 3D video benchmark synthesis (CVPR'26 Oral); Dirichlet-energy latent shaping of VLM 3D-topology subspaces; cross-frame consistency under sequential observation budgets.
- **(Nov 2024 – May 2025)** Drove research on diffusion / generative-model interpretability and personalization: training-free fine-grained natural-language explanations of Stable Diffusion / GAN / autoregressive image generators (FineXL); explainable on-device LLM personalization (MobiSys'25 Xpert).
- **(Sept 2022 – Oct 2024)** Designed federated learning systems for heterogeneous AIoT fleets: gradient correction under data / device heterogeneity (MobiCom'25); federated learning with unbounded device staleness (AAAI'25).

PUBLICATIONS (PEER-REVIEWED)

1. **[CVPR'26 Oral] InfiniBench: Infinite Benchmarking for Visual Spatial Reasoning with Customizable Scene Complexity**
   Haoming Wang, Qiyao Xue, Wei Gao.
   IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026 (Acceptance 25.4%, Oral).
   [arXiv] [Code] [Data]
   - **Architected** a customizable framework that turns plain-language scene descriptions into photo-realistic 3D walk-through videos (Blender Cycles) with controllable identity and layout; established a new evaluation protocol for video-based reasoning – a transferable methodology for long-video quality assessment.
2. **[MobiSys'25] Never Start from Scratch: Expediting On-Device LLM Personalization via Explainable Model Selection**
   Haoming Wang, Boyuan Yang, Xiangyu Yin, Wei Gao.
   Proceedings of the 23rd ACM International Conference on Mobile Systems, Applications and Services (MobiSys), 2025 (Acceptance 18.0%). [Paper]
   - **Developed Xpert**, an explainable model-selection method warm-starting on-device LLM personalization from the closest pre-personalized checkpoint.
3. **[MobiCom'25] When Device Delays Meet Data Heterogeneity in Federated AIoT Applications**
   Haoming Wang, Wei Gao.

Proceedings of the 31st ACM International Conference on Mobile Computing and Networking (MobiCom), 2025 (Acceptance 17.1%). [Paper]

- **Engineered** a federated AIoT training method that jointly disentangles and corrects for data heterogeneity and device delays via targeted gradient correction.

4. **[AAAI'25] Tackling Intertwined Data and Device Heterogeneities in Federated Learning with Unlimited Staleness**
   Haoming Wang, Wei Gao.
   Proceedings of the 39th AAAI Conference on Artificial Intelligence, 2025 (Acceptance 23.4%). [arXiv]
   - **Designed** a gradient-inversion technique that recovers usable training signal from arbitrarily stale device updates, enabling FL to converge on real heterogeneous fleets.

Preprints & Papers Under Review

1. **Deciphering Personalization: Towards Fine-Grained Explainability in Natural Language for Personalized Image Generation Models (FineXL)**
   Haoming Wang, Wei Gao.
   Under Review, 2025. [Paper]
   - **Architected FineXL**, a training-free method decomposing a personalized image generator's divergence from its base into orthogonal natural-language concepts; cut single-aspect explanation error by 56% across **diffusion, GAN, and autoregressive** generators – a new methodology for fine-grained perceptual evaluation of generative-model differences.
2. **FreezeAsGuard: Mitigating Illegal Adaptation of Diffusion Models via Selective Tensor Freezing**
   Kai Huang, **Haoming Wang** (co-author), Wei Gao.
   Preprint, 2024. [Paper] [Code]
   - **Co-developed** a bilevel-optimization defense that learns a tensor-freezing mask over Stable Diffusion – a clean differentiable recipe for fine-grained manipulation of diffusion weights at scale; 37–47% stronger mitigation than unlearning baselines.
3. **Uncovering and Shaping the Latent Representation of 3D Scene Topology in Vision-Language Models**
   Haoming Wang, Wei Gao.
   Preprint, 2026. [Paper] [Code]
   - **Discovered** a VLM latent subspace encoding 3D scene topology and **introduced** a Dirichlet-energy regularizer beating SFT by up to 12.1% – a concrete instance of shaping hierarchical latent structure relevant to extended-generation latent design.
4. **MosaicThinker: On-Device Visual Spatial Reasoning for Embodied AI via Iterative Construction of Space Representation**
   Haoming Wang, Qiyao Xue, Weichen Liu, Wei Gao.
   Under Review, 2026. [arXiv]
   - **Built** a training-free pipeline that stitches multi-view first-person video frames into a compact semantic map; replaced drift-prone sequential alignment with an MST routing scheme anchoring every frame to a global reference – a transferable idea for long-horizon temporal consistency.
5. **Reasoning Path and Latent State Analysis for Multi-view Visual Spatial Reasoning: A Cognitive Science Perspective**
   Qiyao Xue, **Haoming Wang** (co-author), Weichen Liu, Shiqi Wang, Yuyang Wu, Wei Gao.
   Under Review, 2026. [arXiv]
   - **Co-authored** ReMindView-Bench (50K+ VQAs across 15 VLMs); demonstrated internal uncertainty cleanly separates correct vs. incorrect reasoning trajectories – a candidate perceptual / coherence metric for assessing temporal consistency in generated video.
6. **HEVEN & GlobalNav (VLM-based autonomous mobile systems)**
   Anonymous Authors (incl. **Haoming Wang**, 3rd author).
   Double-Blind Under Review, 2026.

- **Co-authored** two VLM-based mobile systems: HEVEN (hierarchical spatial semantic tree + Tree-of-Thought planner; 73.86% success on a real quadruped) and GlobalNav (infrastructure-assisted navigation, 90.91% success in simulation).

7. **Spatial Reasoning in Multimodal Large Language Models: A Survey of Tasks, Benchmarks and Methods**
   Weichen Liu, Qiyao Xue, **Haoming Wang**, Xiangyu Yin, Boyuan Yang, Wei Gao.
   Under Review, 2026.

HONORS & AWARDS

April 2026 **ECE Research Assistant of the Year**
Department of Electrical and Computer Engineering, University of Pittsburgh

TECHNICAL SKILLS

Generative Models: Stable Diffusion, ControlGAN, Anole-7B, CLIP / FG-CLIP; bilevel mask optimization, linear-representation decomposition, Laplacian-eigenmap latent probing.

Video / 3D Pipelines: Blender Cycles, Infinigen, Habitat, Isaac Sim; frontier-based camera planning, MatchAnything, DepthPro, SAM-3D / 3DGS.

Programming: Python, PyTorch, CUDA, Hugging Face, ONNX, TensorRT.